

SHIVAM SINGH [SHIV]

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PRINCIPAL TECHNICAL PROGRAM MANAGER — DIGITAL SOVEREIGNTY, GENERATIVE AI & E-COMMERCE

EDUCATION

University of Michigan, Stephen M. Ross School of Business — Master of Business Administration | 2022–2024 | Ann Arbor, MI

Concentrations in **Strategy, Technology & Operations**; Zell Lurie Institute for Entrepreneurial Studies

The London School of Economics and Political Science — Executive Coursework in Economics, Markets & Public Policy | 2024 | London, UK

Exchange term focused on **capital markets, regulatory economics, and quantitative policy analysis**

Indian Institute of Technology — Bachelor of Technology, Computer Science & Engineering | 2010–2014

Peer-reviewed publication, **Journal of Cleaner Production (Elsevier)**, 159 citations; distributed systems, algorithms, machine learning

PROFESSIONAL EXPERIENCE

AWS — Principal Technical Program Manager, Generative AI & Sovereign Cloud | 2024–Present | San Francisco, CA

Agentic GenAI Shopping Copilot — Consumer E-Commerce (100M+ Users)

- Owned program strategy and cross-functional delivery for an **agentic GenAI shopping copilot** spanning **12 workstreams across 6 orgs** (engineering, applied science, legal, trust & safety, GTM, localization) in **3 time zones**: hardened the **RAG pipeline**, tuned **prompt engineering and model selection** on **Amazon Bedrock**, and configured **Bedrock Guardrails, evals, and human-in-the-loop** controls for high-risk responses.
- Drove adoption to **25%**, brand favorability **55% → 70%**, hallucination rate to **2.8%**, and session depth **+15%**; authored the **Working Backwards / PR-FAQ** executive narrative and **\$100M+ GMV** sensitivity model that secured **\$5M** funding from senior leadership against competing bets.
- Shipped **Japan localization (i18n)** — tokenization and honorific handling — as the **reusable multi-market expansion template**, with a work-back schedule, risk register, and launch-readiness reviews coordinating **8+ teams**.

Sovereign Cloud GTM — Regulated Banking Sector

- Architected a **sovereign-cloud go-to-market** for systemically important banks, translating **data-sovereignty, operator-access, and resiliency** requirements into reference architectures and **compliance-as-code** deployment guardrails; unlocked **\$20M+** in stalled cloud commitments, built a **\$122M** pipeline, and cut compliant deployment **6 months → 2 hours** — versus the industry-standard manual accreditation cycle.

Rakuten — MBA Intern, E-Commerce Strategy & Marketplace Programs | 2023 | Tokyo, Japan

Marketplace GTM Realignment — B2B2C Growth & Unit Economics

- Stress-tested a proposed **\$50M** DTC plan via a **10,000-iteration Monte Carlo simulation** of CAC, churn, and payback distributions; presented the downside-tail evidence that led leadership to kill the plan and pivot the marketplace to a **B2B2C corporate-partnership model**, aligning category, finance, and tech stakeholders behind the new business case.
- Pivot delivered **CAC -80% (\$42 → \$8.50)**, **LTV:CAC 0.8 → 4.5**, **D30 retention 34% → 67%**, and payback **18 months → 4 months**, validated with **geo-lift testing and cohort analytics (SQL)**.

Microsoft — Senior Program Manager, Regulated Industries & Cross-Border Compliance | 2020–2022 | Redmond, WA

Secure AI Platform for Regulated Financial Services — Compliance as a Feature

- Translated regulatory requirements into technical architecture for AI fintechs on Azure: executed a **fully-homomorphic-encryption → Azure Confidential Computing (secure enclaves)** pivot (inference latency **14s → 220ms**), deployed **Azure AI Content Safety guardrails** and a **bias-testing sandbox**, and codified the rollout as **SOPs** adopted across **10+** customer engagements.
- Cut **time-to-compliance 9 months → 3 months**, sustained **zero breaches** (~\$6M/company risk preserved), and lifted **AI ROI +61%** — making compliance a **go-to-market accelerator** rather than a blocker, versus the industry norm of compliance as a launch gate.

Cross-Border Data Compliance & FinTech Cloud Architecture (10 Markets)

- Rescued a **\$5M** at-risk regulated account and won a **\$2M** upsell (**\$7M** total): halted a non-compliant launch over **\$25M** fine exposure, then led the **data-residency re-architecture** across **10 markets** (**GDPR / Schrems II, PIPL, SEC 17a-4**) with engineering and legal — building the **escalation management, risk register, and compliance-by-design SOPs** that became the team standard.

The D. E. Shaw Group — Vice President, Real-Time Platforms & Regulatory Technology | 2016–2020 | New York, NY & Hyderabad, India

Firm-Wide Real-Time Risk Platform under Prudential Regulation

- Led a firm-wide real-time risk and capital-efficiency program — **15+ engineers, 5 stakeholder groups** (trading, quant research, compliance, IT, finance), **\$10M** capex — built on **Rust, Kafka, Flink, and FPGA** under **Basel III** constraints, using a hard **latency budget, Strangler-Fig migration, and shadow-mode validation (90-day parallel run, 98% output agreement)** before cutover.
- Cut risk latency **90ms → 4.2ms**, unlocked **\$85M** in regulatory capital, reduced false positives **15% → 1.8%** (~\$400K/month saved), and scaled throughput to **5.4M msg/sec** with **zero dropped packets** through live market crashes.

McKinsey & Company — Engagement Associate, Retail & Digital Strategy | 2014–2016 | London, UK

APAC Market Entry via Regulatory Engineering & Data Modernization

- Enabled a **\$1.1B** APAC market entry for a financial-services client by pairing **data-platform modernization** with **regulatory engineering** (**Vietnam Decree 53** data localization, **Indonesia OJK** reverse-solicitation), translating regulator constraints into architecture and GTM choices for the **CFO, CIO, and CMO**.

Retail Pricing Strategy & Change Management

- Designed **elasticity-based retail pricing (log-log regression, difference-in-differences)** with a **KVI vs long-tail** portfolio and **Agile Pods**; delivered a **\$50M** revenue unlock, speed-to-market **6 weeks → <1 week**, and **95%** forecast accuracy, winning over a skeptical Chief Merchant via **hindcasting**.

CORE COMPETENCIES

Technical Program Management · program portfolio, risk & escalation management · work-back schedules, launch readiness, risk registers · **SOPs, KPIs & SLAs** · **Digital Sovereignty & Data Residency** · EU regulations & certifications (**GDPR / Schrems II, SecNumCloud, C5, PIPL, SEC 17a-4**) · **regulatory-to-technical translation** · **Generative & Agentic AI** (LLMs, RAG, prompt engineering, guardrails, evals) · **Working Backwards / PR-FAQ** & narratives · influence without authority · executive & VP-level engagement · e-commerce & retail strategy · business case & unit economics · cross-functional delivery (engineering, science, GTM) · supply chain & delivery experience · **SQL & analytics** · consulting (McKinsey)